

- 1 Polyphenol oxidase is an enzyme involved in the browning of fruit.

Fig. 1.1 shows browning of the skin of bananas.

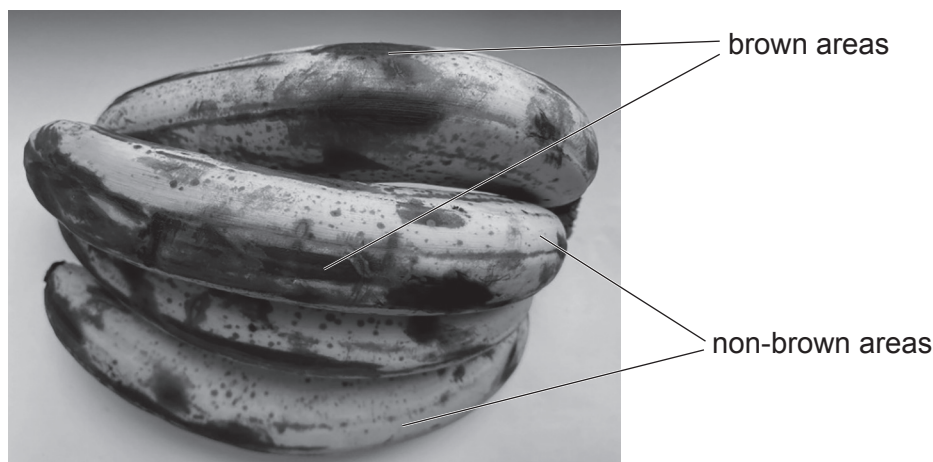


Fig. 1.1

A student extracted polyphenol oxidase enzyme from ripened bananas. The student used this enzyme extract to investigate the factors affecting the activity of the enzyme.

The student initially found that the enzyme extract was too concentrated, so the student made a 1:10 dilution of the enzyme extract.

- (a) Describe how the student made a 1:10 dilution of the enzyme extract.

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..... [1]

- (b) The student studied the effect of pH on the activity of polyphenol oxidase enzyme.

Dopamine hydrochloride solution was used as a substrate for the enzyme. The product of this reaction is red in colour.

A colorimeter can be used to follow the progress of this enzyme-catalysed reaction.

The student set up:

- 7 colorimeter tubes containing a solution of the substrate, dopamine hydrochloride
- 6 colorimeter tubes containing diluted polyphenol oxidase extract, each in a different buffered solution: pH 3.0, pH 4.0, pH 5.0, pH 6.0, pH 7.0, pH 8.0
- 1 colorimeter tube containing water in pH 7.0 buffer.

The colorimeter tubes were equilibrated at a set temperature.

A colorimeter was set up with a filter transmitting light at 470 nm.

At time 0 seconds, the substrate solution and the polyphenol oxidase solution at pH 3.0 were mixed in one colorimeter tube. The colorimeter tube was then immediately put into the colorimeter.

The absorbance was measured at 15 second intervals over the first three minutes of the reaction.

The procedure was repeated for the remaining buffered solutions and the water at pH 7.0.

(i) Identify the **independent** variable in this experiment.

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(ii) Identify **one** variable that the student has standardised.

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..... [1]

- (c) The student then plotted a graph of the results, showing the increase in absorbance. Absorbance is measured in absorbance units (au).

Fig. 1.2 shows the results for the increase in absorbance at **pH 5.0**.

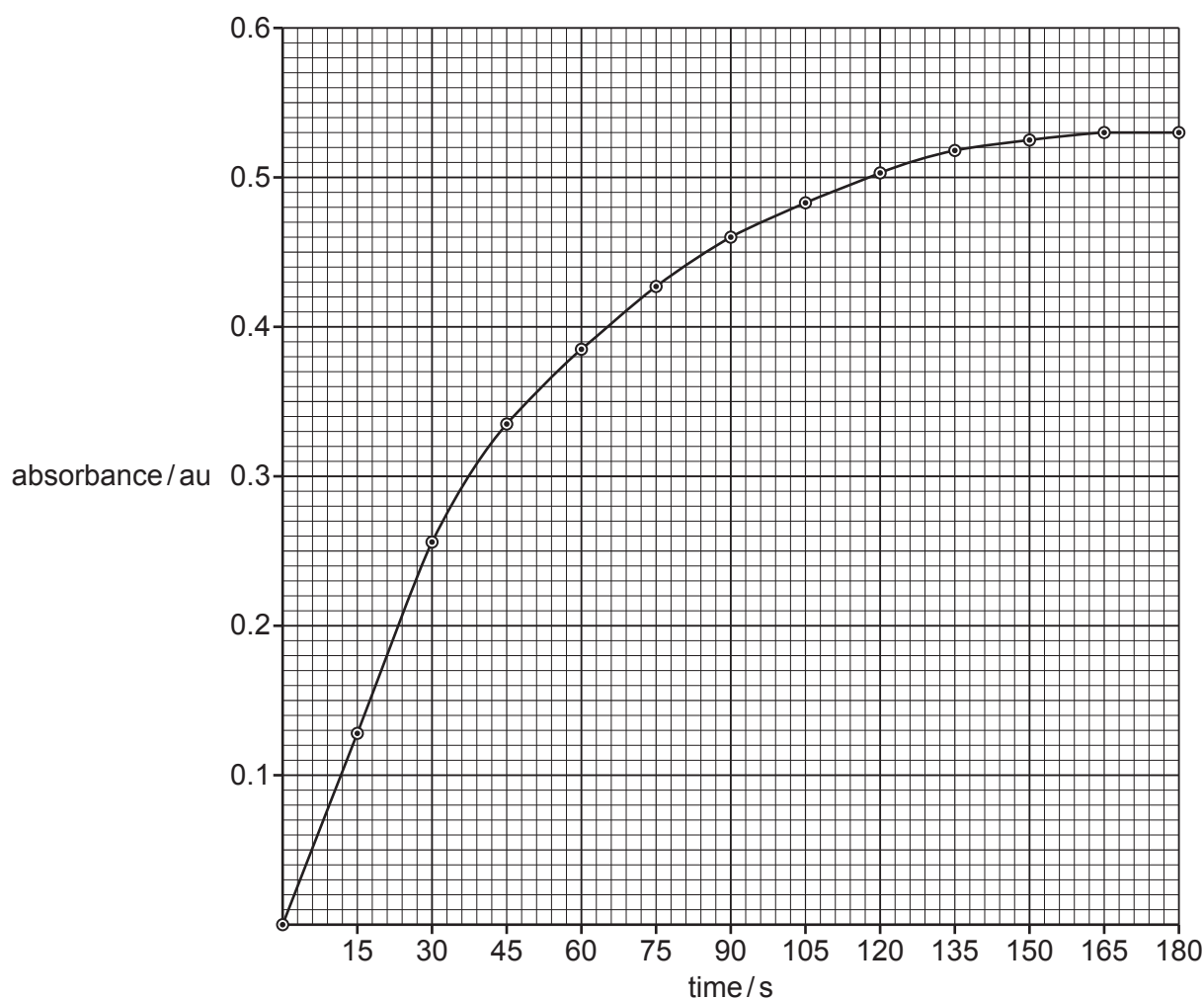


Fig. 1.2

- (i) Use Fig. 1.2 to calculate the initial rate of this reaction in absorbance units per second (au s^{-1}).

Show your working.

initial rate of reaction = au s^{-1} [2]

- (ii) Using the internet, the student found that pH 7.0 is the optimum pH for polyphenol oxidase.

On Fig. 1.2, sketch the line you would expect to get for the rate of reaction at **pH 3.0**. [1]

- (d)** The student wanted to determine the effect of the concentration of the substrate, dopamine hydrochloride solution, on the rate of the enzyme-catalysed reaction.

The student was provided with a stock solution of 0.2% dopamine hydrochloride solution and the diluted enzyme extract prepared in **(a)**.

Describe a method the student could use to collect the data that is needed to determine the effect of the concentration of dopamine hydrochloride solution on the rate of the enzyme-catalysed reaction.

Your method should be set out in a logical order and be detailed enough to let another person follow it.

[illegible]

- (e) The browning of fruit such as bananas can make them more difficult to sell. One way to prevent browning of bananas is to use an anti-browning agent.

The student searched the internet and found data from different scientific papers on the inhibitory effect of various anti-browning agents on polyphenol oxidase from different plants.

The student processed and ranked these data, as shown in Table 1.1.

Table 1.1

anti-browning agent	percentage of maximum activity of polyphenol oxidase	plant source of polyphenol oxidase
heated onion extract	17.0	potato
fresh onion extract	44.4	potato
ascorbic acid	60.0	sorrel
L-cysteine	70.0	sorrel
potassium sorbate	76.8	potato
citric acid	80.4	potato
EDTA	88.0	red poppy
no anti-browning agent (control)	100.0	potato

The student concluded that heated onion extract was the most effective anti-browning agent and that EDTA was the least effective.

Use the information to suggest why this conclusion may **not** be supported.

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- (f) The student found that a sulfur-containing molecule in the heated onion extract acted as a non-competitive inhibitor to the enzyme polyphenol oxidase.

Fig. 1.3 is a graph showing how the substrate concentration affects the rate of polyphenol oxidase activity.

On Fig. 1.3 sketch the curve that shows the rate of activity for polyphenol oxidase treated with the heated onion extract.

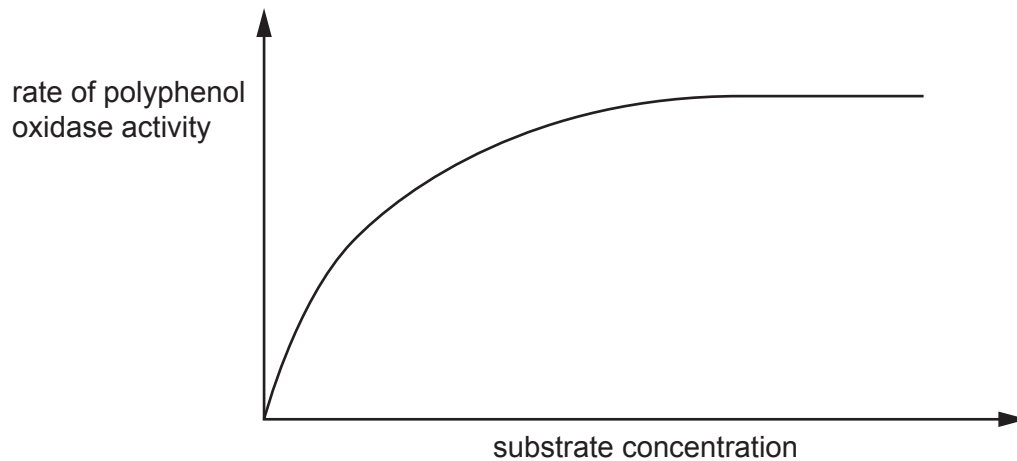


Fig. 1.3

[1]

[Total: 16]